AHMED HANNAN

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PROFESSIONAL SUMMARY

Software Engineer and Computer Science graduate (FAST-NUCES) with a proven record of shipping concurrent, production web and mobile platforms alongside cutting-edge research in Multimodal Generative AI and High-Performance Computing. At EmpowerBits, engineered production web and mobile apps (iOS & Android) actively supporting concurrent users, earning an executive Certificate of Excellence / Performance Award. Experienced in building end-to-end multimodal RAG pipelines (CLIP, FAISS, Gemma 3), GPU-accelerated computing (CUDA, Tensor Cores), distributed graph algorithms (OpenMP, MPI), and cloud analytics (Google BigQuery). Demonstrates strong architectural ownership spanning full-stack application development, systems engineering, and applied AI research.

TECHNICAL COMPETENCIES

Languages: Python, C/C++ (Modern / STL), C#, Dart, JavaScript (ES6+), TypeScript, SQL, CUDA C, x86 Assembly, HTML5/CSS3

AI & Deep Learning: PyTorch, Hugging Face Transformers, Gemma 3, CLIP, FAISS Vector Search, Cross-Encoder Reranking, YOLOv8, OpenCV, PixelCNN, LSTM/RNN

RAG & LLM Orchestration: Multimodal RAG, RAG-Fusion, HyDE, Corrective RAG (CRAG), Graph RAG, LangChain, LangGraph, Model Context Protocol (MCP)

Full-Stack & Mobile: React.js, Next.js, Flutter (iOS & Android), Node.js, Express.js, Django REST Framework, Flask, WebSockets, Tailwind CSS, Vite

Systems, Cloud & Tools: CUDA Kernels, Tensor Cores, OpenMP, MPI, Google BigQuery, PostgreSQL, MongoDB, Redis, Docker, Git/GitHub, Linux/Bash, Selenium, Jira

PROFESSIONAL EXPERIENCE

EmpowerBits | Full-Stack Developer June 2026 — Present

- **Production Applications:** Architected, developed, and deployed production-ready applications with active concurrent user traffic, spanning high-availability web platforms and cross-platform mobile apps for iOS and Android.
- **Performance Optimization:** Implemented reactive state management in Flutter and React, coupled with optimized backend microservices and client-side caching to maintain sub-100ms response times under heavy concurrent loads.
- **Recognition:** Awarded official Certificate of Excellence / Performance Award for exceeding technical delivery milestones and taking end-to-end architectural ownership of client-critical features.
- Tech Stack: Flutter, Dart, React.js, Node.js, Python, PostgreSQL, REST APIs, WebSockets, Git.

RedCoast | Full-Stack Software Engineer June 2025 — June 2026

- **Client Delivery:** Spearheaded full-stack feature delivery across multiple client applications, translating complex business specifications into scalable software architecture.
- **Data Pipelines:** Built automated data extraction and transformation pipelines integrated with Google BigQuery for real-time analytical querying and enterprise dashboards.
- **Quality Engineering:** Designed automated end-to-end regression suites using Selenium WebDriver and integrated continuous quality tracking within Jira CI/CD workflows, cutting query latency by 35%.
- Tech Stack: React.js, Flutter, Django REST Framework, Google BigQuery, PostgreSQL, Selenium, Jira, Git.

National Center of Artificial Intelligence (NCAI, NUST) | AI Research Intern August 2024 — September 2024

- **Generative AI Research:** Investigated architectures of generative deep learning models including autoencoders, diffusion principles, and vision-language transformers for cross-modal alignment.
- **Model Benchmarking:** Trained and benchmarked deep transformer models across distributed GPU clusters, evaluating zero-shot retrieval accuracy against standard computer vision datasets.
- Tech Stack: Python, PyTorch, Hugging Face, Vision Transformers, CUDA, Linux GPU Clusters.

FAST — NUCES | Undergraduate Teaching Assistant (Computer Science) June 2024 — September 2024

- **Mentorship:** Conducted lab tutorials and code review sessions for 100+ undergraduate students covering Data Structures & Algorithms and Programming Fundamentals in C++.
- **Systems Foundation:** Mentored students on dynamic memory allocation, pointer manipulation, cache locality, and asymptotic algorithmic complexity analysis.

EDUCATION

FAST — National University of Computer and Emerging Sciences (NUCES) Graduation: June 2026

Bachelor of Science in Computer Science | Islamabad, Pakistan

Core Coursework: Distributed Systems, High-Performance Computing, Deep Learning, Generative AI, Computer Vision, Operating Systems, Design & Analysis of Algorithms, Database Systems, Computer Networks.

KEY ENGINEERING & RESEARCH PROJECTS

Pose2Play: Real-Time AI Computer Vision & Unity VR Platform (Final Year Project) GitHub: Pose2Play

- **Architecture:** Engineered an interactive VR rehabilitation system integrating computer vision pose estimation via YOLOv8 with real-time Unity VR gameplay.
- **Performance:** Streams 3D skeletal joint kinematics via low-latency inter-process communication (IPC) for sub-30ms exercise posture correction and rep counting.
- Tech Stack: Python, YOLOv8, OpenCV, C#, Unity VR, PyTorch, ZeroMQ.

Multimodal RAG & Gemma-3 Semantic Product Search Suite GitHub: multimodal-rag-semantic-search

- **Engine Architecture:** Constructed an end-to-end multimodal retrieval-augmented generation engine powered by Google Gemma-3-1b-it FP16 inference with layer offloading for constrained VRAM environments.
- **Reranking & Retrieval:** Integrated hybrid FAISS indexing (CLIP visual embeddings + MiniLM text embeddings) with fine-tuned BERT-Tiny cross-encoder reranking on 1.1M Amazon ESCI queries.
- Tech Stack: PyTorch, Hugging Face, FAISS, CLIP, Gemma 3, Cross-Encoder, Streamlit, Python.

CRAG: Advanced Multi-Pipeline RAG Benchmark & Dashboard GitHub: crag-advanced-rag-pipelines

- **Retrieval Suite:** Implemented and benchmarked 7 retrieval architectures (RAG Fusion with Reciprocal Rank Fusion, HyDE, Corrective RAG with web fallback, Graph RAG, RRR) on the Meta CRAG dataset.
- **Full-Stack Deployment:** Developed full-stack evaluation dashboard featuring a Flask REST API backend, global vector indexing, and an interactive React + Vite visualization interface.
- Tech Stack: Python, LangChain, Meta CRAG, FAISS, Flask, React.js, Vite, spaCy.

CUDA Deep Learning Kernel Acceleration GitHub: Neural-Network-Acceleration

- **HPC Engineering:** Implemented forward and backward neural network propagation routines and General Matrix Multiply (GEMM) operations from scratch in CUDA C++.
- **Kernel Optimization:** Utilized 2D shared memory block tiling, warp-level primitives, and Tensor Cores, achieving a 12x speedup over standard multi-threaded CPU routines.
- Tech Stack: CUDA C++, NVIDIA Tensor Cores, GPU Shared Memory, C++, Linux.

ParButterfly: Parallel Billion-Edge Graph Motif Mining GitHub: ParButterfly

- **High-Performance Computing:** Engineered a scalable parallel algorithm for butterfly-motif counting on massive billion-edge bipartite networks using OpenMP shared-memory and MPI distributed computing.
- **Scalability:** Optimized vertex sorting order, thread cache locality, and atomic synchronization, delivering near-linear multi-core scalability.
- Tech Stack: C/C++, OpenMP, MPI, High-Performance Computing, Graph Theory.

GoMatch: Real-Time Geospatial Ride-Sharing Mobile Platform GitHub: GoMatch

- **Real-Time Mobile:** Developed an on-demand ride sharing mobile application built with Flutter/Dart and a Django REST API backend with PostGIS spatial database queries.
- **Telemetry:** Engineered live driver-rider geolocation matching, bidirectional WebSocket coordinate telemetry streaming, automated fare estimation, and interactive routing.
- Tech Stack: Flutter, Dart, Django REST Framework, WebSockets, PostgreSQL, PostGIS, Google Maps API.

QuizVerse: Interactive Full-Stack Assessment Platform GitHub: QuizVerse

- **Full-Stack MERN:** Architected a responsive examination management platform featuring Google OAuth 2.0 and JWT authentication, role-based access control, timer engines, and live analytics.
- Tech Stack: React.js, Node.js, Express.js, MongoDB, OAuth 2.0, JWT, REST APIs.

HONORS & ACHIEVEMENTS

- Executive Certificate of Excellence / Performance Award — EmpowerBits (2026): Awarded for exceptional technical velocity and architectural delivery across concurrent production web and mobile apps.
- Competitive Teaching Assistantship Appointment — FAST-NUCES (2024): Selected on academic merit to mentor 100+ undergraduate computer science students in core programming and data structures.